Core-collapse supernova and host demographics in the Dark Energy Survey

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February 2022

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Overview

Supernovae Background

- Introduce type Ia and core-collapse supernovae
- Discuss why core-collapse supernovae are of interest
- Detail how we can study them

Core-collapse Supernovae in DES

- Introduce the Dark Energy Survey (DES)
- Detail methods to calculate supernova and host properties
- Present results

Generating Synthetic Supernovae

- Introduce Generative Adversarial Networks (GANs)
- Explain use of GANs for generating supernova light curves
- Present intermediate results

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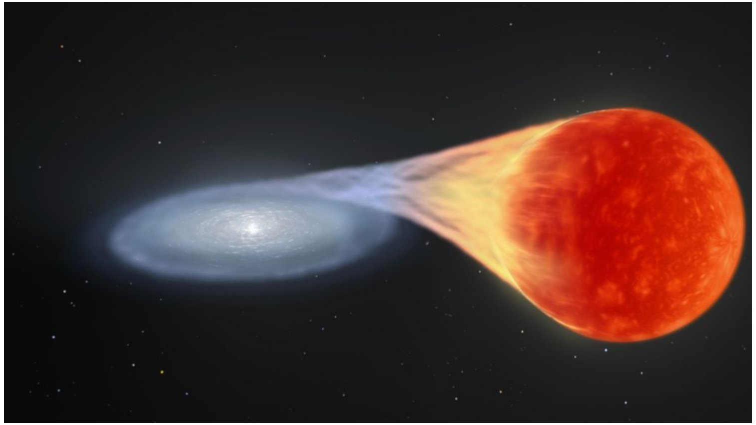
Supernovae

Type Ia

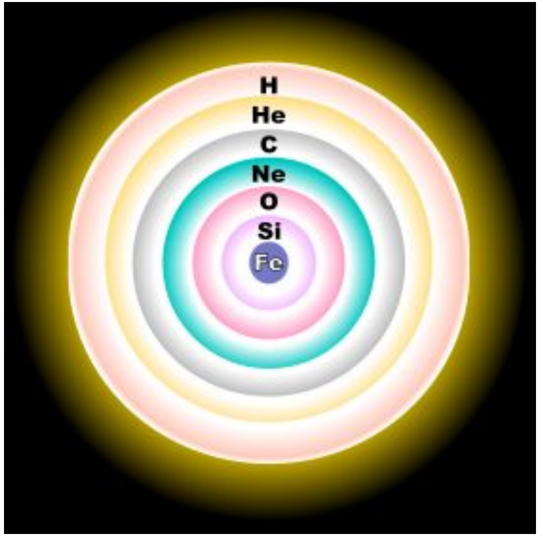
- White dwarf progenitor
- Standardisable candles



Credit: Adriana Manrique Gutierrez svs.gsfc.nasa.gov

Core-collapse

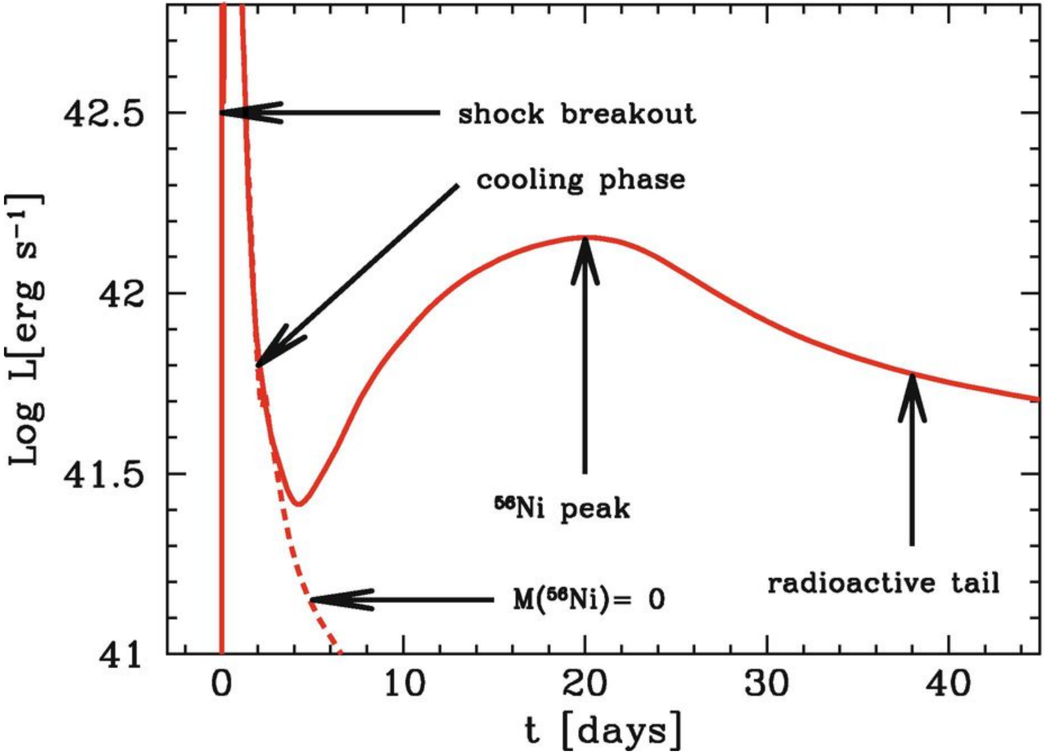
- Massive star progenitor
- Very diverse properties



Why are Core-collapse Supernovae Interesting?

- A very common type of cosmic explosion, but a great deal not understood about the underlying physics
- Create and distribute high-mass elements
- Significant effect on galaxies, injecting energy and expelling gas

Timeline of a core-collapse supernova



Bersten+2011

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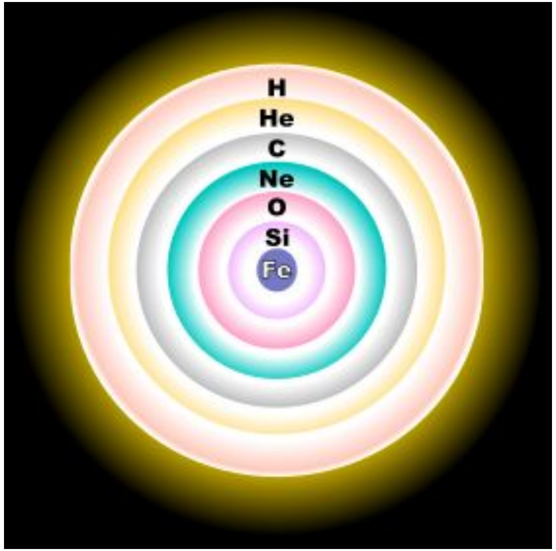
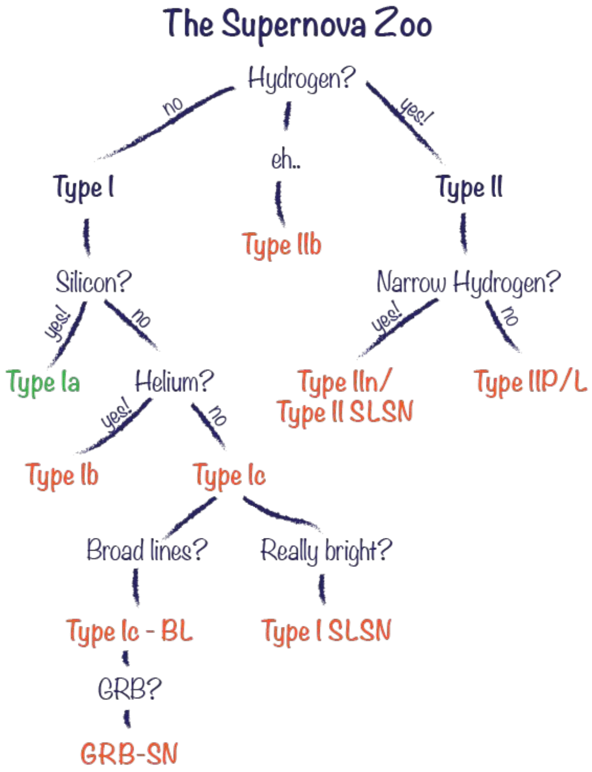
Core-collapse subtypes

- For this talk, two broad categories
  - Type II/H-rich
  - Type Ibc/H-poor

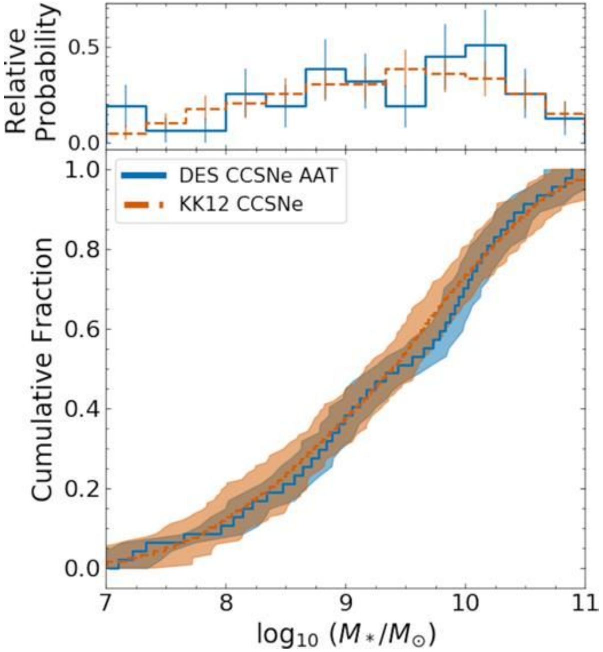




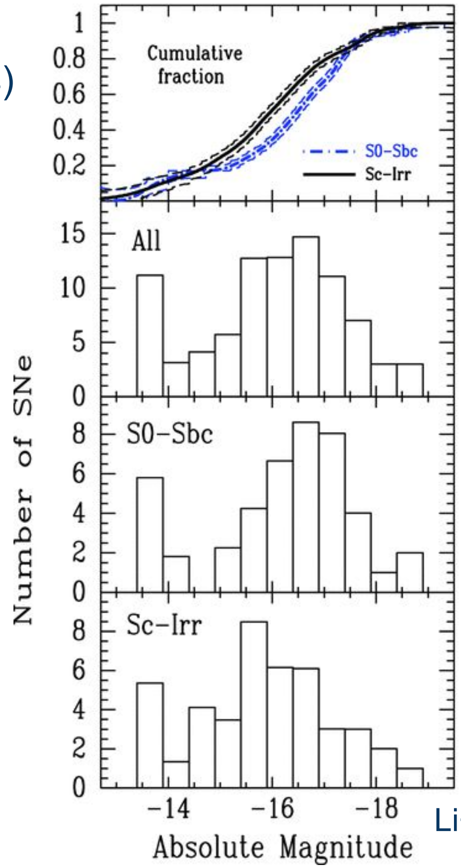
Credit: Ashley Villar

How can we study them?

- Luminosity functions (Distribution of peak luminosities)
- Host galaxy properties



Wiseman+20



Li+2011

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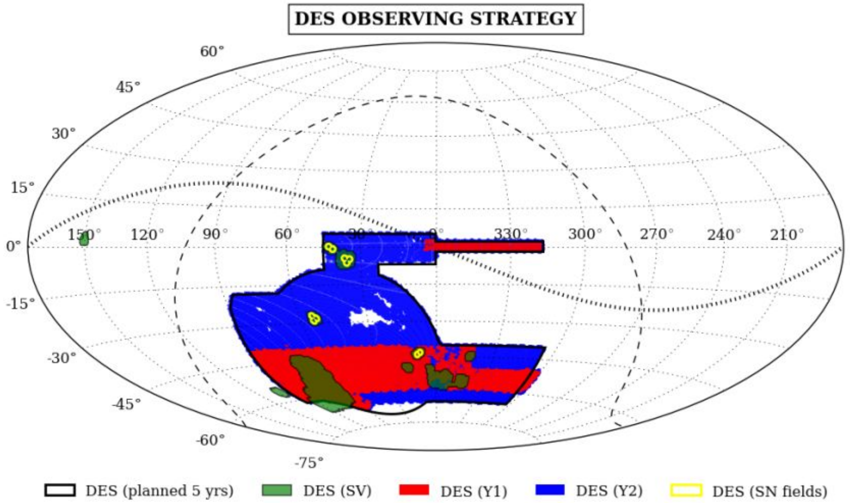
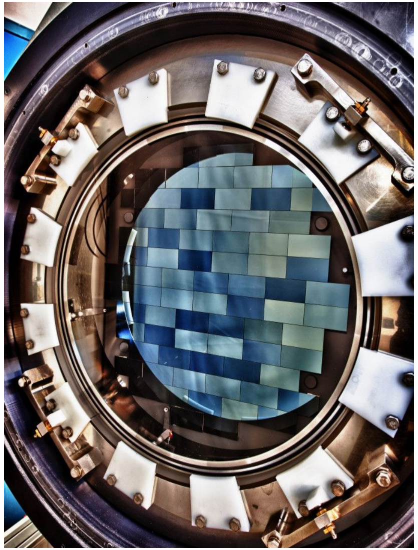
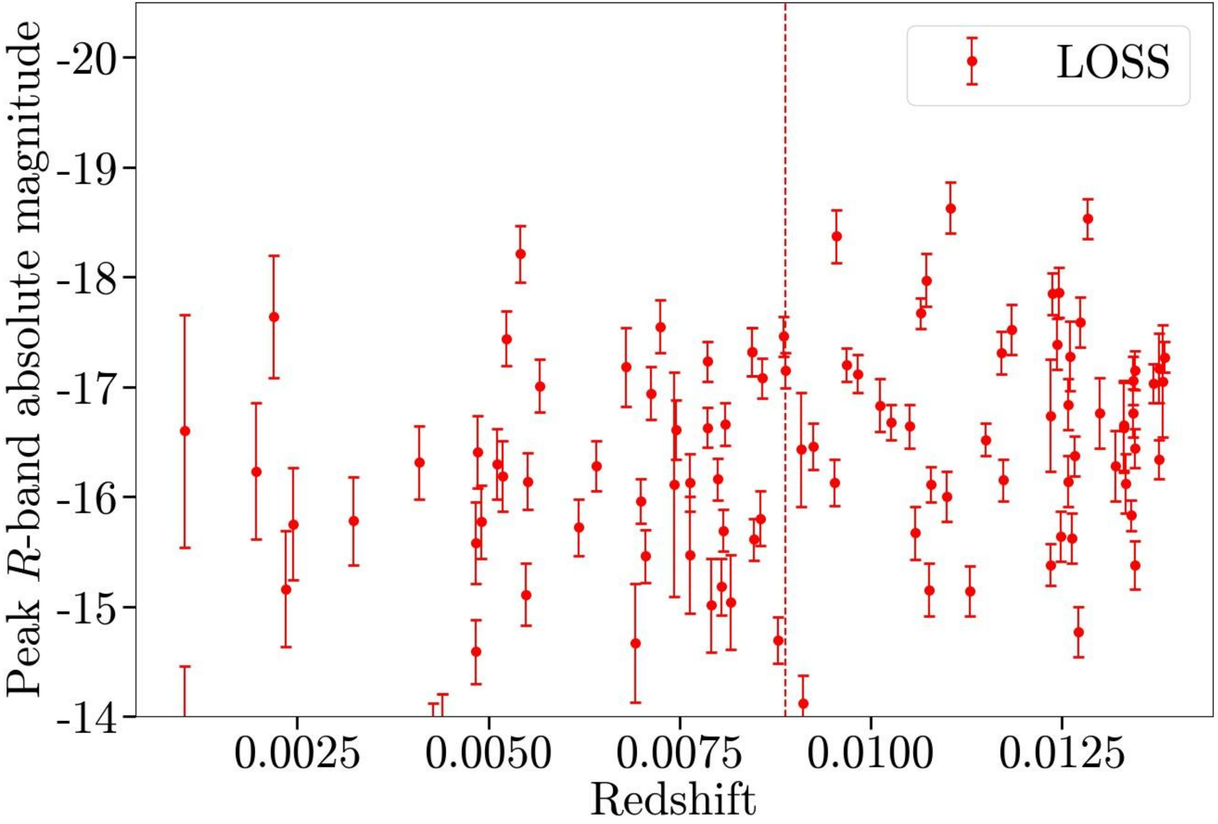
The Dark Energy Survey (DES)

- Aim - “Everything with one survey”, studying dark energy using multiple probes
- DES-SN
  - Observed 27 square degrees in *griz* bands every 7 nights for 5 years
  - Deep (~24 mag), untargeted rolling SN search





Why DES?


LOSS
Peak R-band absolute magnitude
-20
-19
-18
-17
-16
-15
-14
0.0025
0.0050
0.0075
0.0100
0.0125
Redshift

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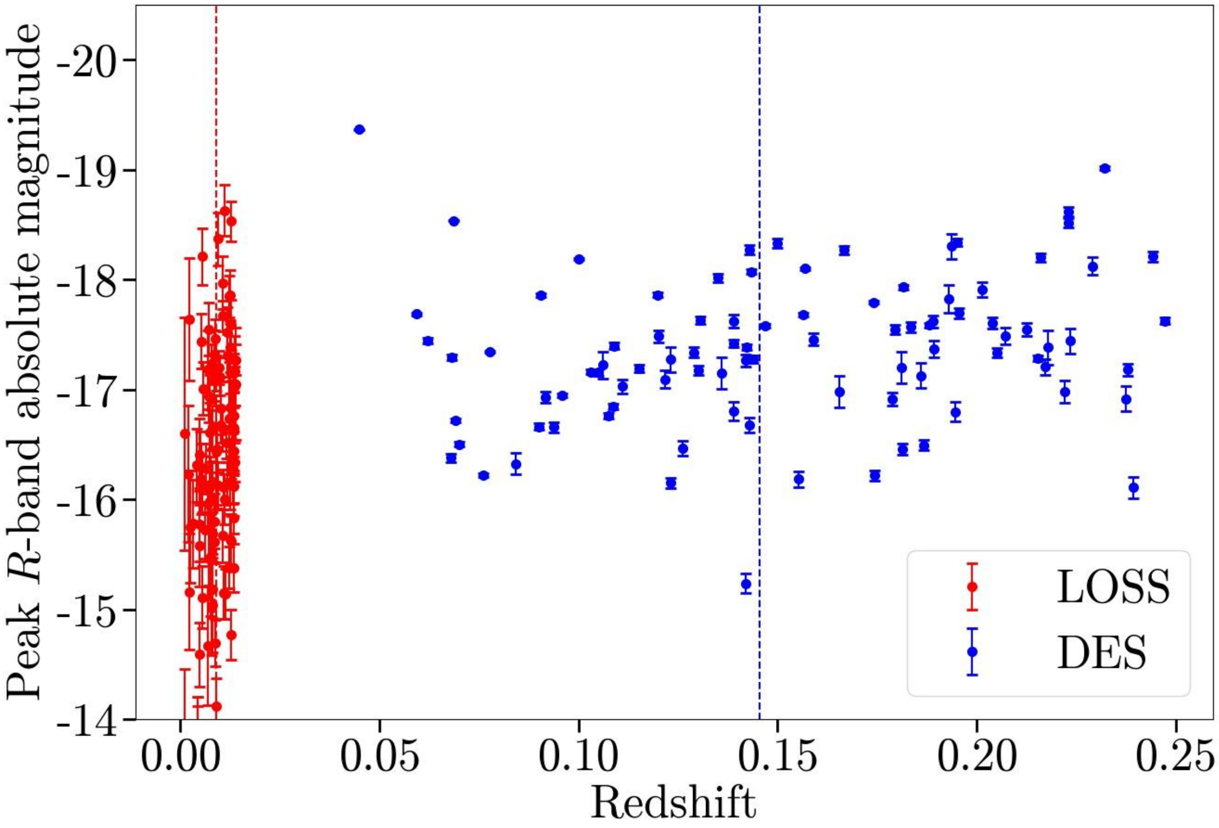
Why DES?



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DES Core-collapse Supernova Samples

- Spectroscopically confirmed
  - Classified using SN spectrum
  - 52 SNe II, 18 SNe Ibc
- Spectroscopic host
  - Classified using pSNiD
  - 51 SNe II, 22 SNe Ibc
- Photometric host
  - No spectroscopic host redshift, only photometric redshift

Method: Luminosity Functions

1. Apply quality cuts
2. Estimate peak luminosity:
   a. Correct for MW extinction
   b. Gaussian process (GP) interpolate to obtain simultaneous observations
   c. Use SED models from simulations to k-correct to rest frame
   d. GP interpolate again to obtain peak luminosity in rest-frame
3. Correct for Malmquist bias using V_{max} correction:

$$w = \begin{cases} \left(\dfrac{d_c\left(z_{\mathrm{survey}}\right)}{d_c\left(z_{\mathrm{max}}\right)}\right)^3 & \text{if } z_{\mathrm{max}} < z_{\mathrm{survey}} \\ 1 & \text{otherwise} \end{cases}$$

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Core-Collapse Supernova Samples

- Dark Energy Survey (DES)
  - 0.045 < z < 0.25 (High redshift by CCSN standards!)
  - 61 SNe II, 30 SNe Ibc
- Lick Observatory Supernova Search (LOSS)
  - Very local, z < 0.014
  - Galaxy targeted
  - 43 SNe II, 25 SNe Ibc
- Zwicky Transient Facility (ZTF)
  - Between DES and LOSS, 0.005 < z < 0.06
  - Rolling, untargeted survey
  - 174 SNe II, 90 SNe Ibc

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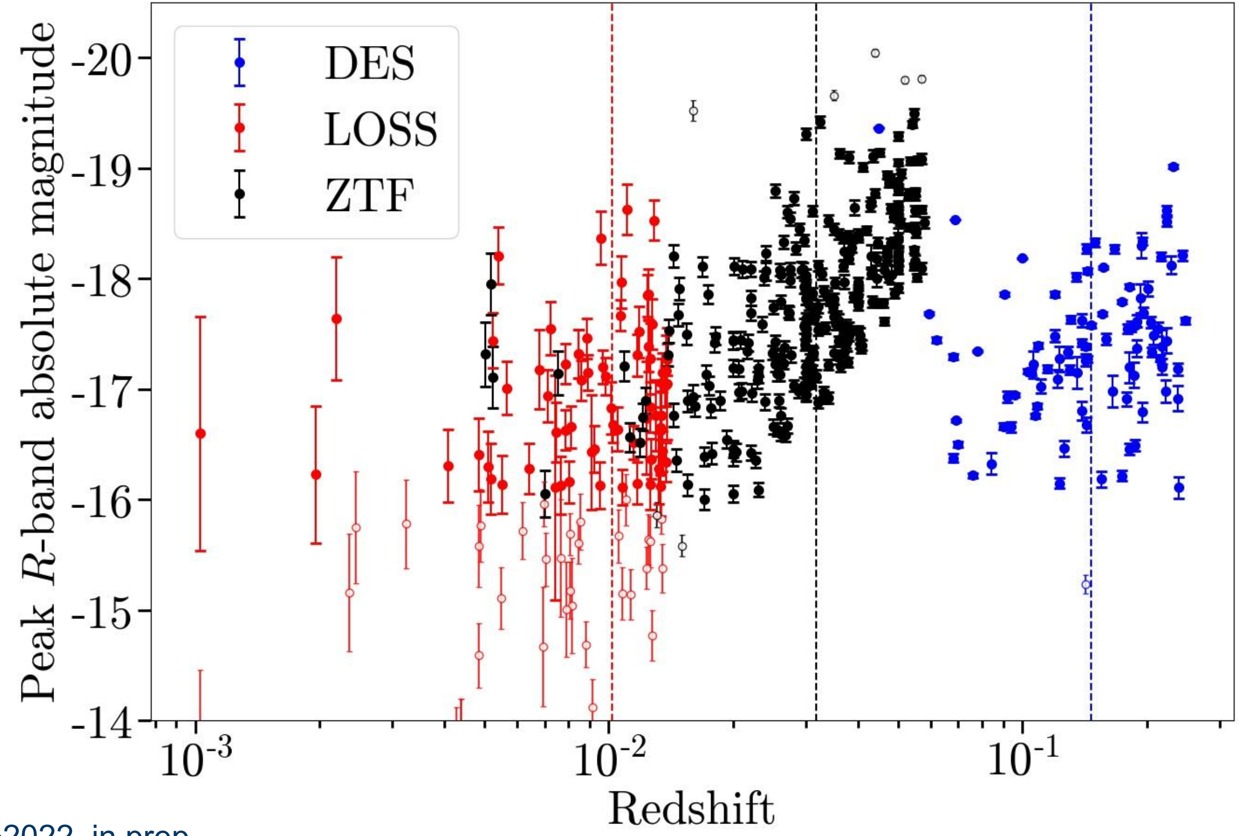
Comparison Samples



Grayling+2022, in prep.

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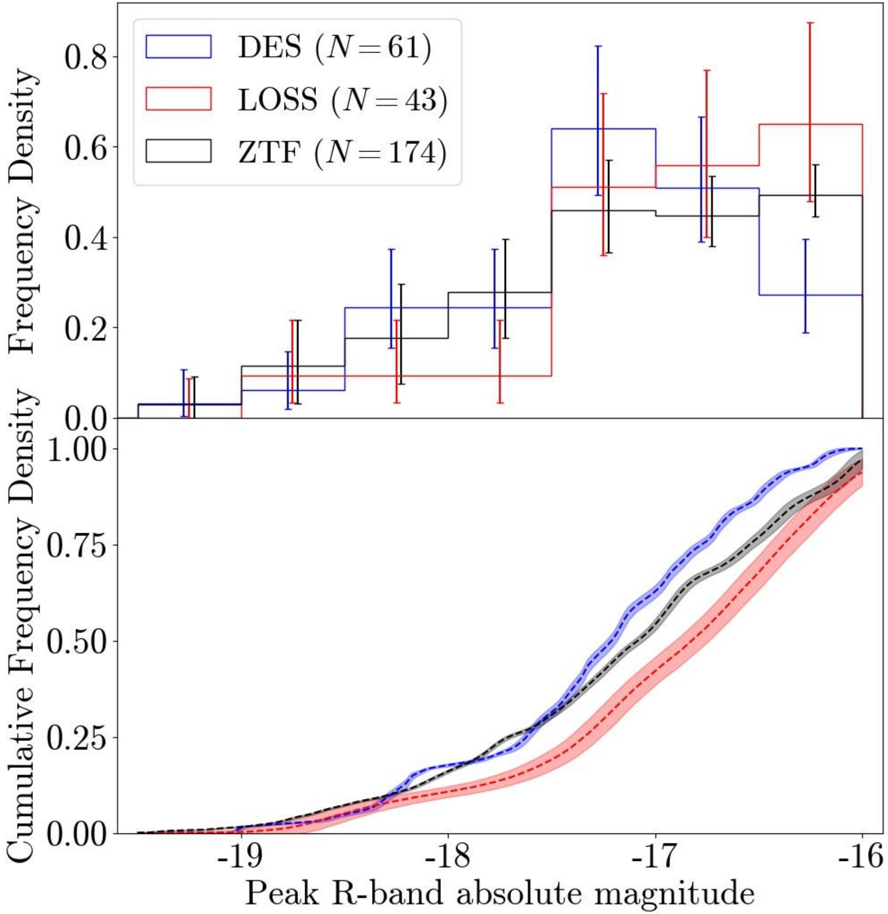
Luminosity Functions - SNe II



DES/LOSS: 3.0σ
LOSS/ZTF: 2.1σ

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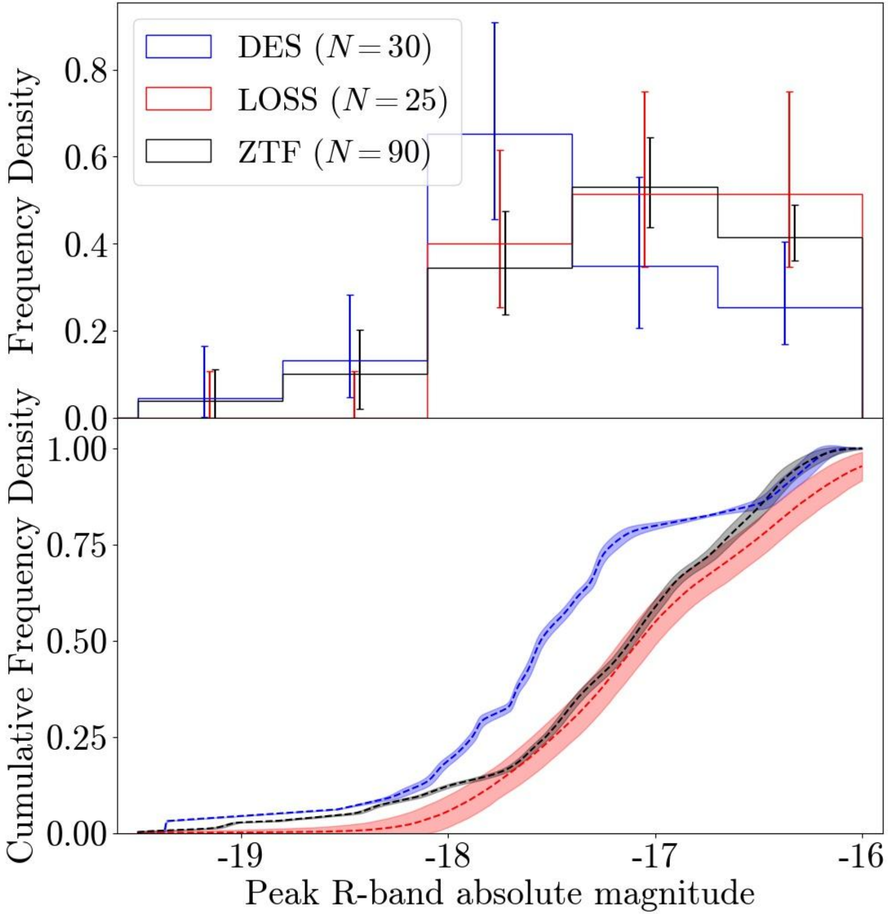
Luminosity Functions - SNe Ibc



DES/LOSS: 2.5σ

DES/ZTF: 2.7σ

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Differences in the Luminosity Function?

- Some evidence to suggest differences in luminosity functions between DES, LOSS and ZTF
- Could indicate a redshift evolution in the luminosity function
- However, could be a result of simpler effects such as differing levels of host extinction

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Method: Host Galaxy Properties

- Can calculate host properties from SED fits to galaxy photometry
- We use *griz* photometry for DES from deep co-added photometry from Wiseman+2020, and *ugriz* photometry from SDSS for LOSS and ZTF
- SED fits carried outs using PEGASE.2 templates and a Kroupa IMF
- Main focuses:
    - Stellar mass
    - Rest-frame U-R colour (proxy for star formation)

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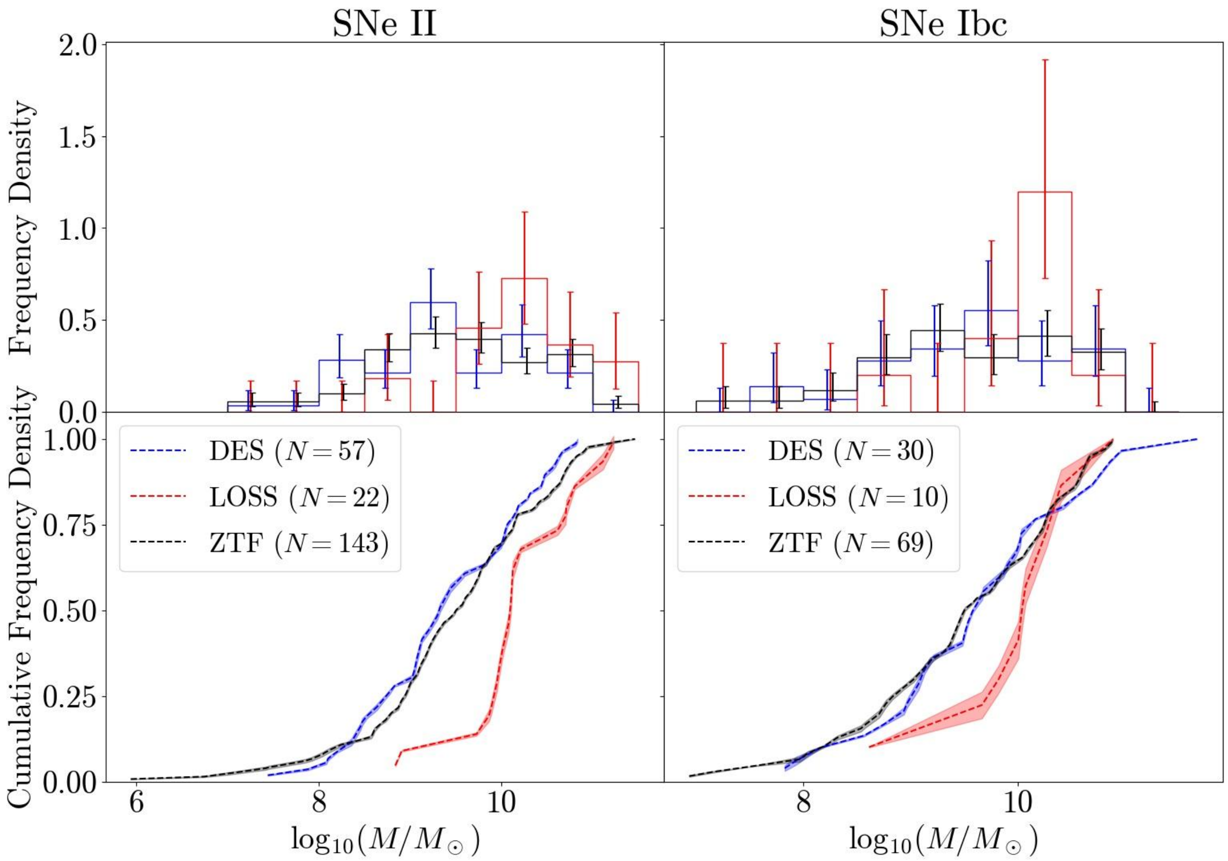
Host Galaxy Properties - Stellar Mass



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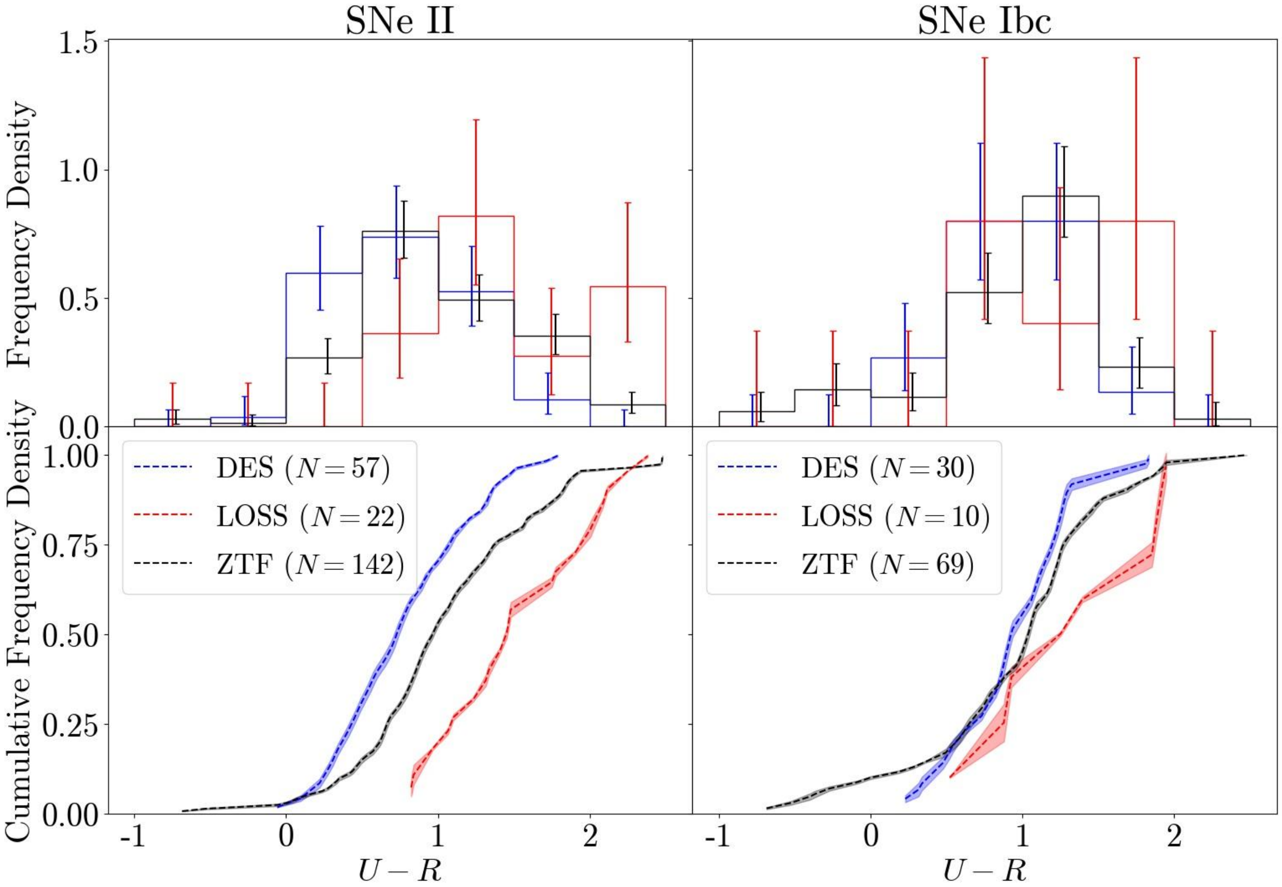
Host Galaxy Properties - U-R Colour



DES/ZTF: 2.8σ

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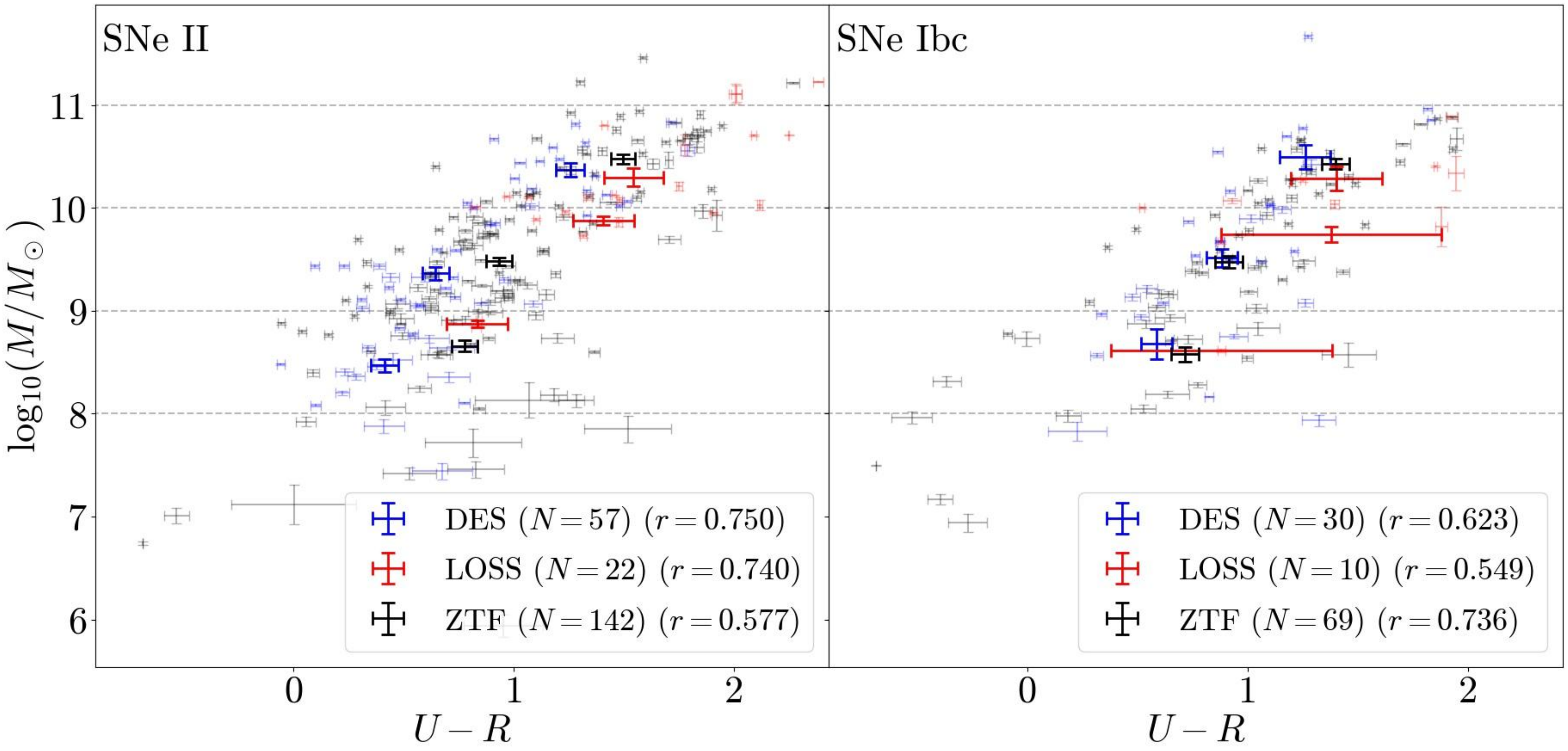
Host Galaxy Properties - U-R vs Mass



What causes the host colour difference?

- Redshift evolution?
    - We know that galaxies evolve with redshift, this could explain the difference

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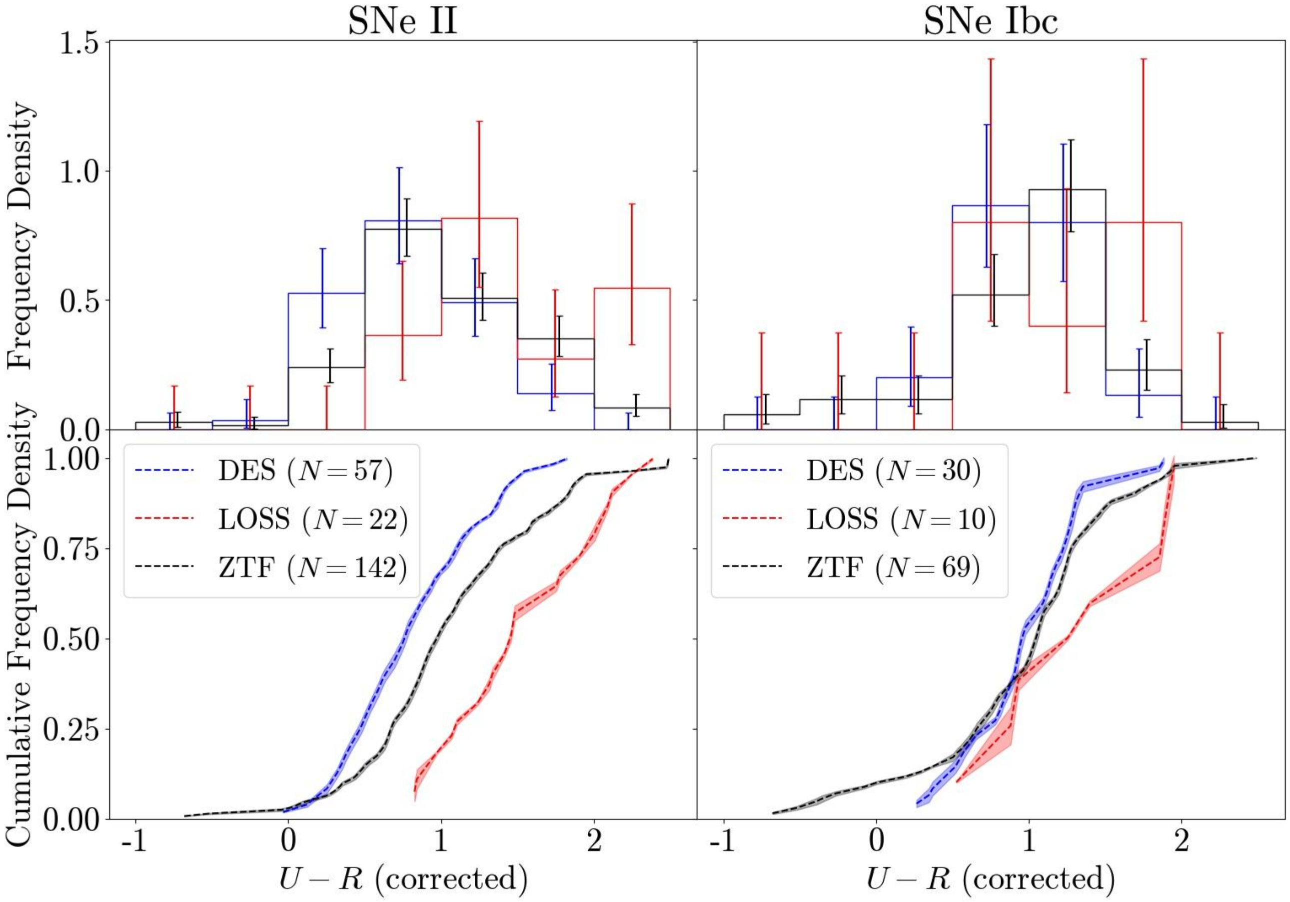
Host Galaxy Properties - U-R Colour (corrected)



DES/ZTF: 2.7σ

What causes the host colour difference?

- Not redshift evolution
- Selection effects?
  - We need spectroscopic host galaxy redshift
  - Measured from emission lines which are stronger in bluer galaxies, maybe there is a bias?

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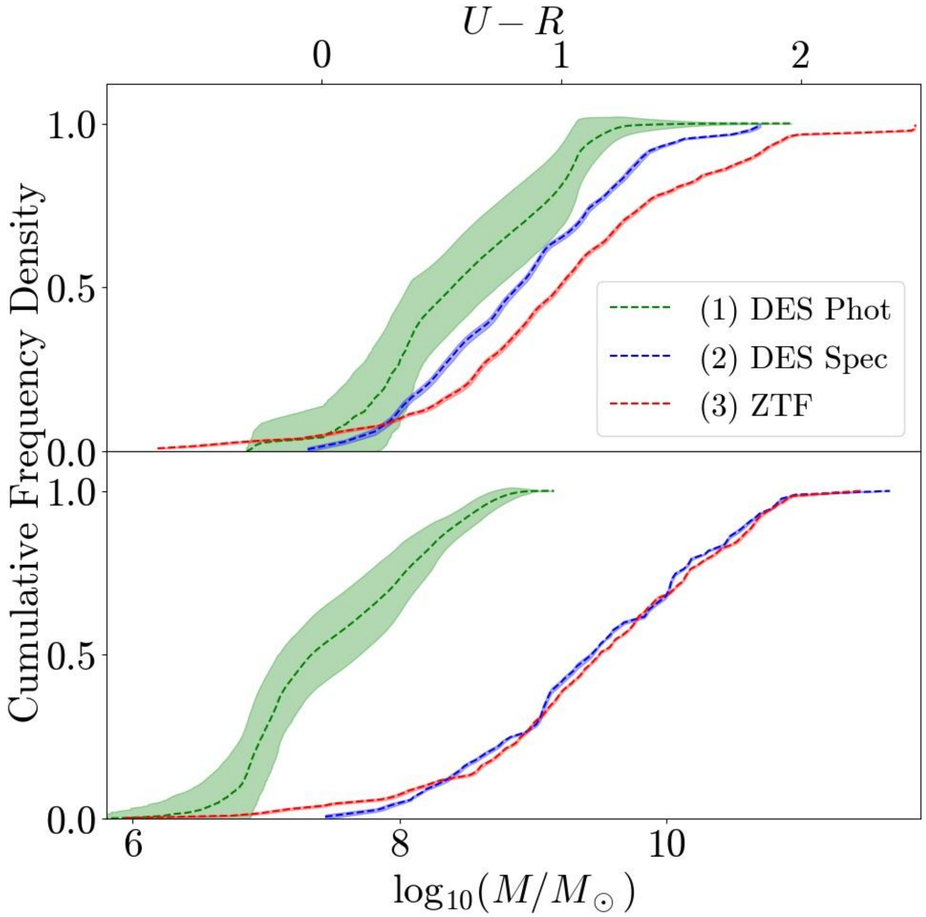
Host Galaxy Properties - Photometric Redshift



Not a DES selection bias

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What causes the host colour difference?

- Not redshift evolution
- Not selection effects
- Systematic offset?
  - We use SDSS *ugriz* host photometry for ZTF but DES *griz* photometry, maybe this causes an offset?

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What causes the host colour difference?

- Not redshift evolution
- Not selection effects
- Not systematic offset
- ?

Photometric Classification

- Surveys such as LSST will observe far more objects than we are able to obtain spectra for
- Good progress made in photometrically classifying SNe Ia, but classifying core-collapse SNe can be trickier due to the abundance of varied types
- A lack of supernovae of rarer classes is an obstacle to classifying these objects well

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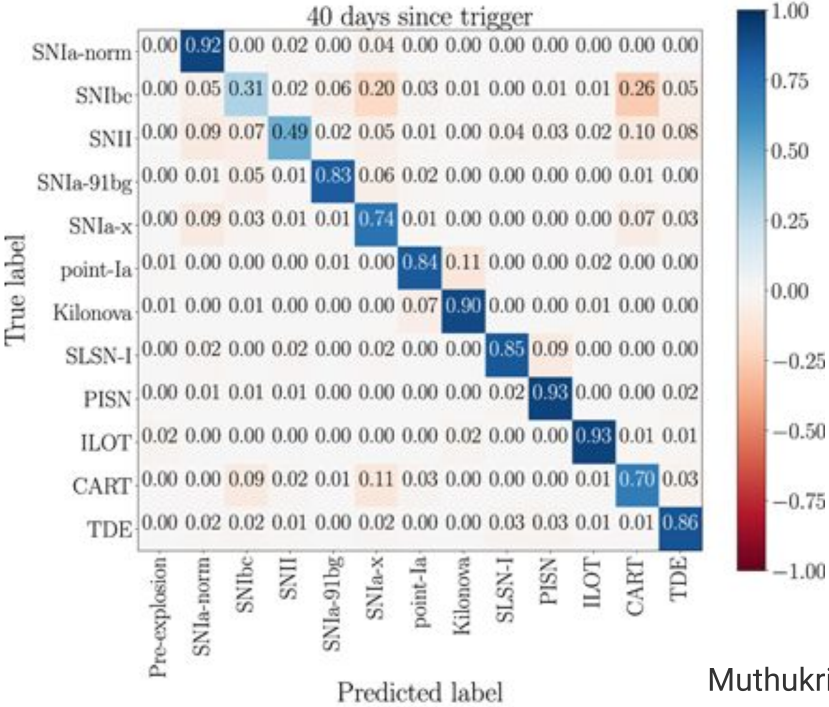
Photometric Classification



Muthukrishna+2019

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Data Generation

Credit: https://thispersondoesnotexist.com/

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Data Generation

Credit: https://thiscatdoesnotexist.com/

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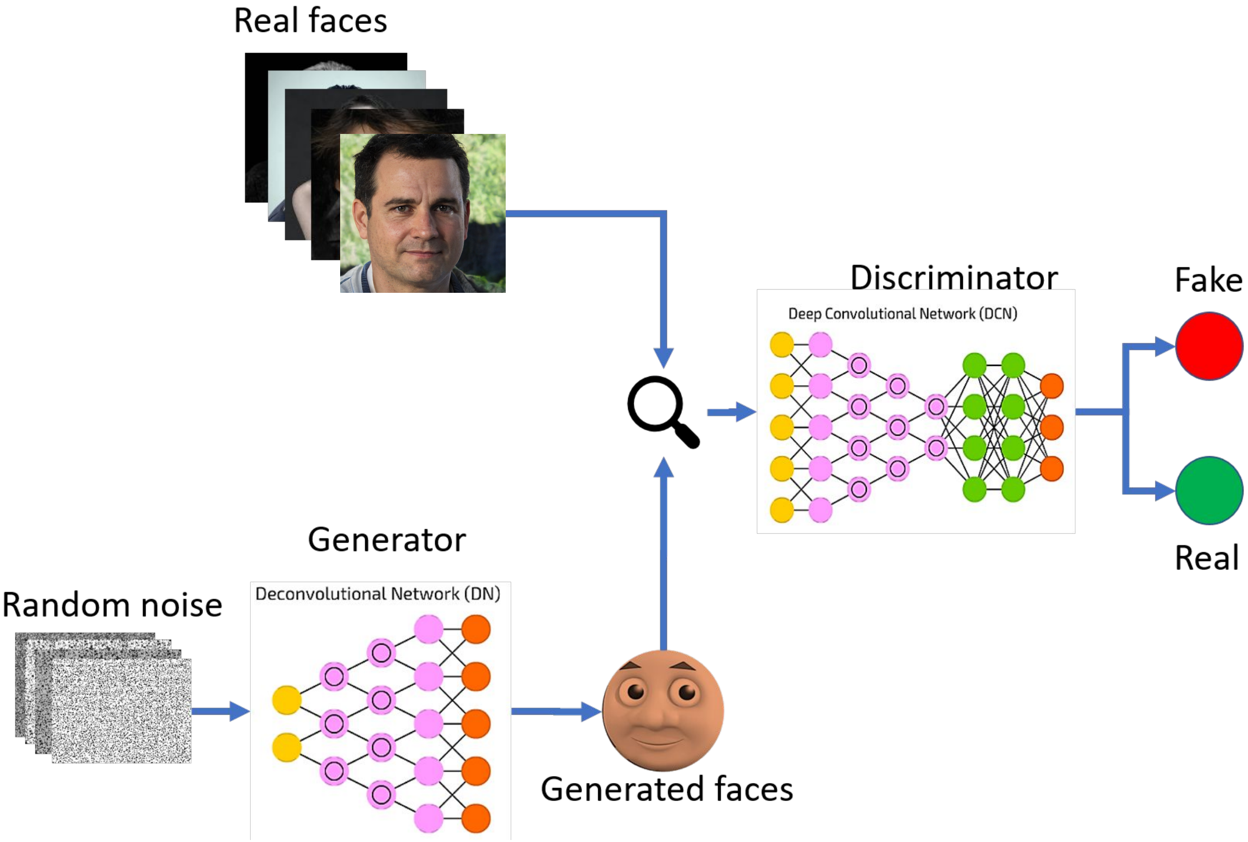
Generative Adversarial Networks (GANs)



Aim: Generate Synthetic Supernovae

- Can generate synthetic data to boost sample sizes
- This approach has been used in fields such as medicine to improve classification performance
- No assumptions need to be made about underlying Physics of explosion, will learn from full range of observed light curves

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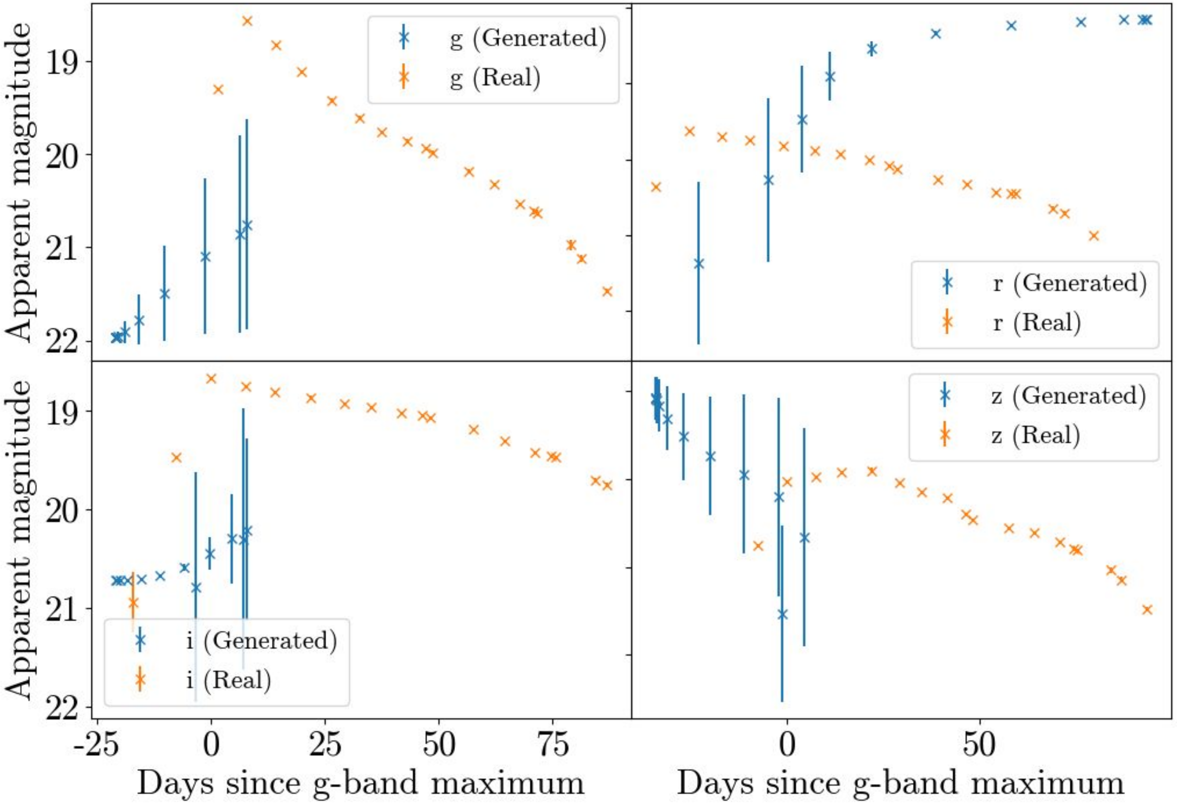
Core-collapse Supernovae in DES

Generating Synthetic Supernovae

Training: Epoch 1



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Core-collapse Supernovae in DES

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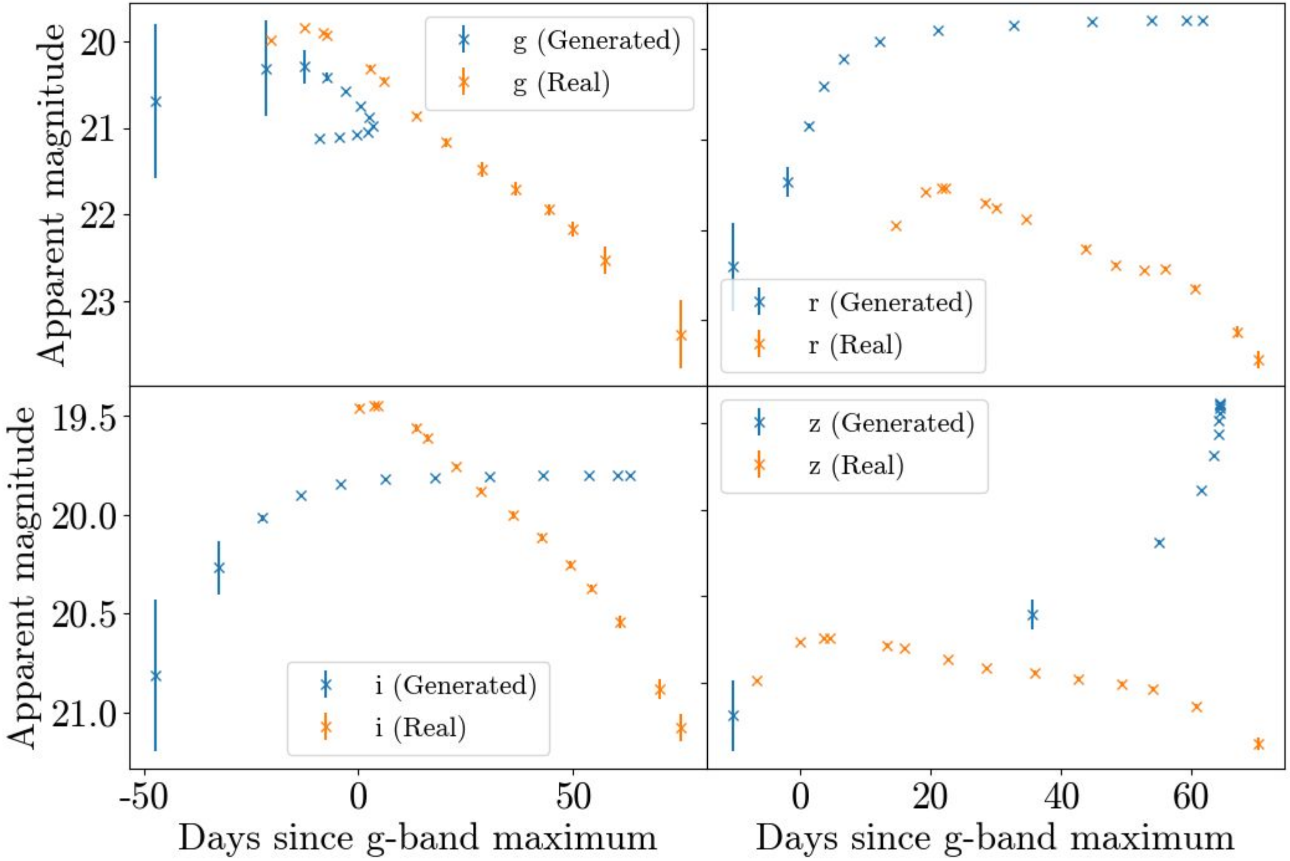
Training: Epoch 5



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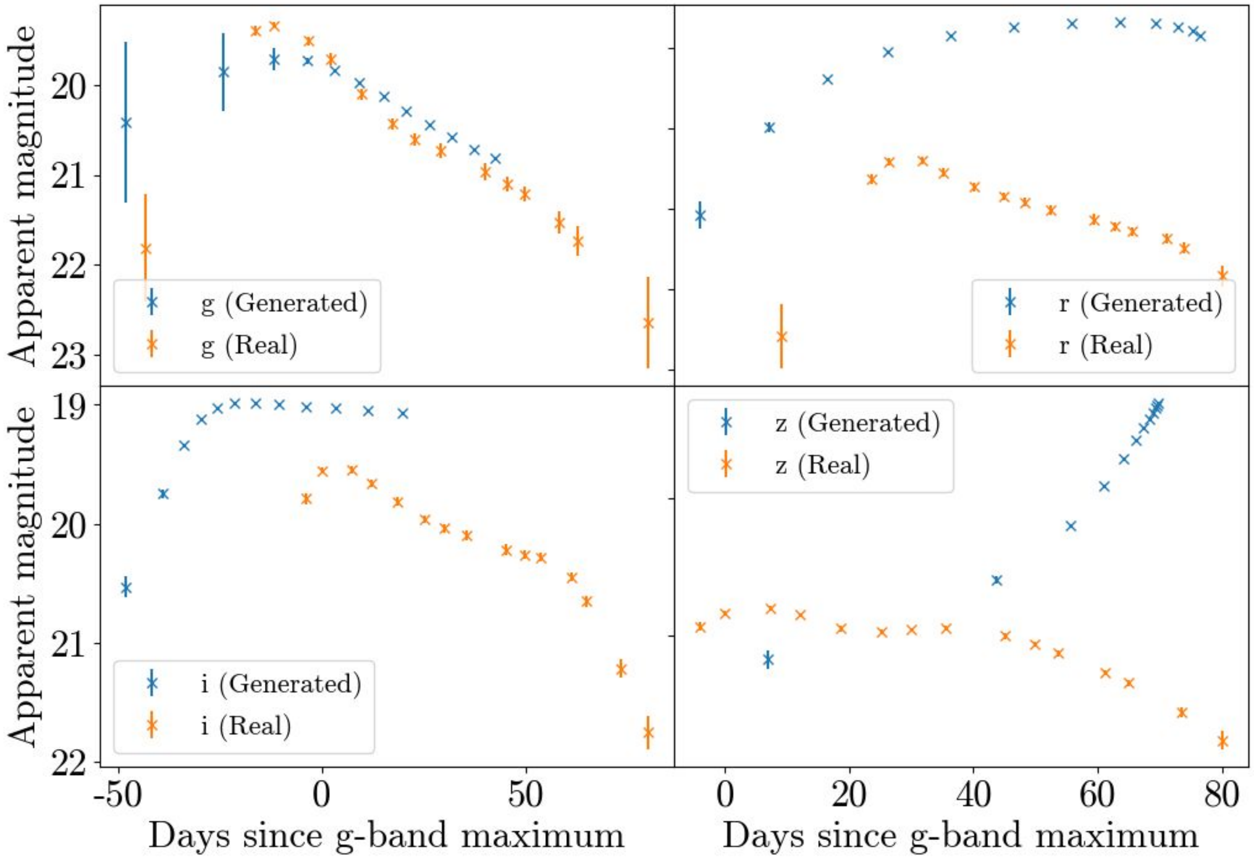
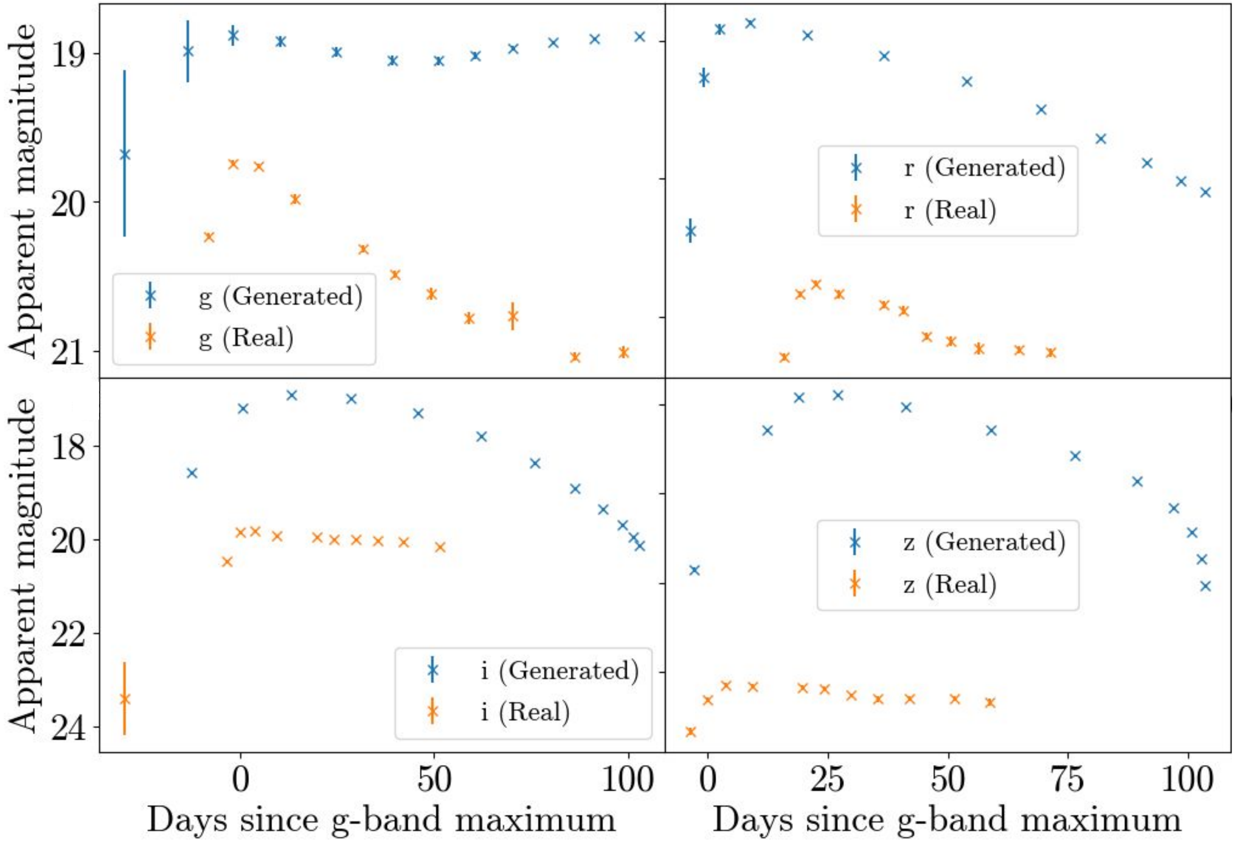
Training: Epoch 10



Training: Epoch 30



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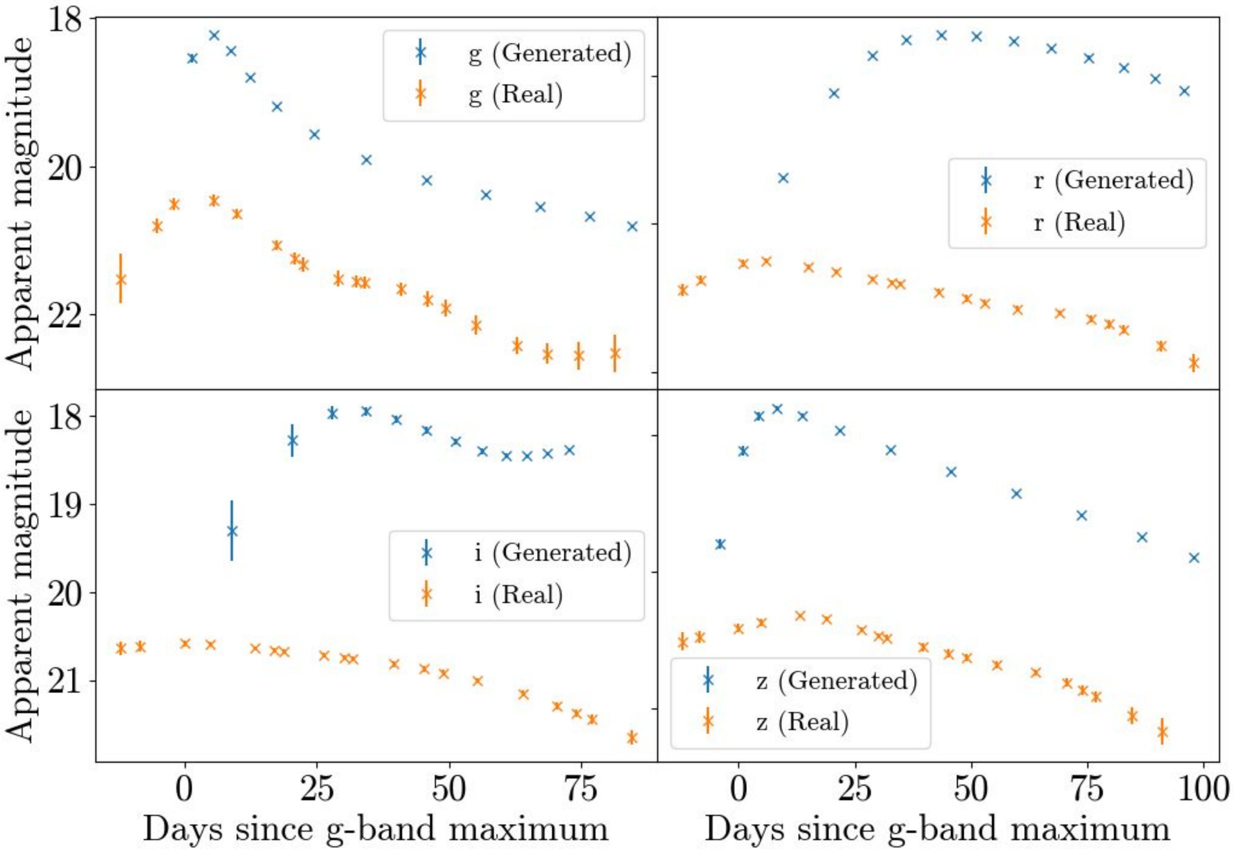
Training: Epoch 80



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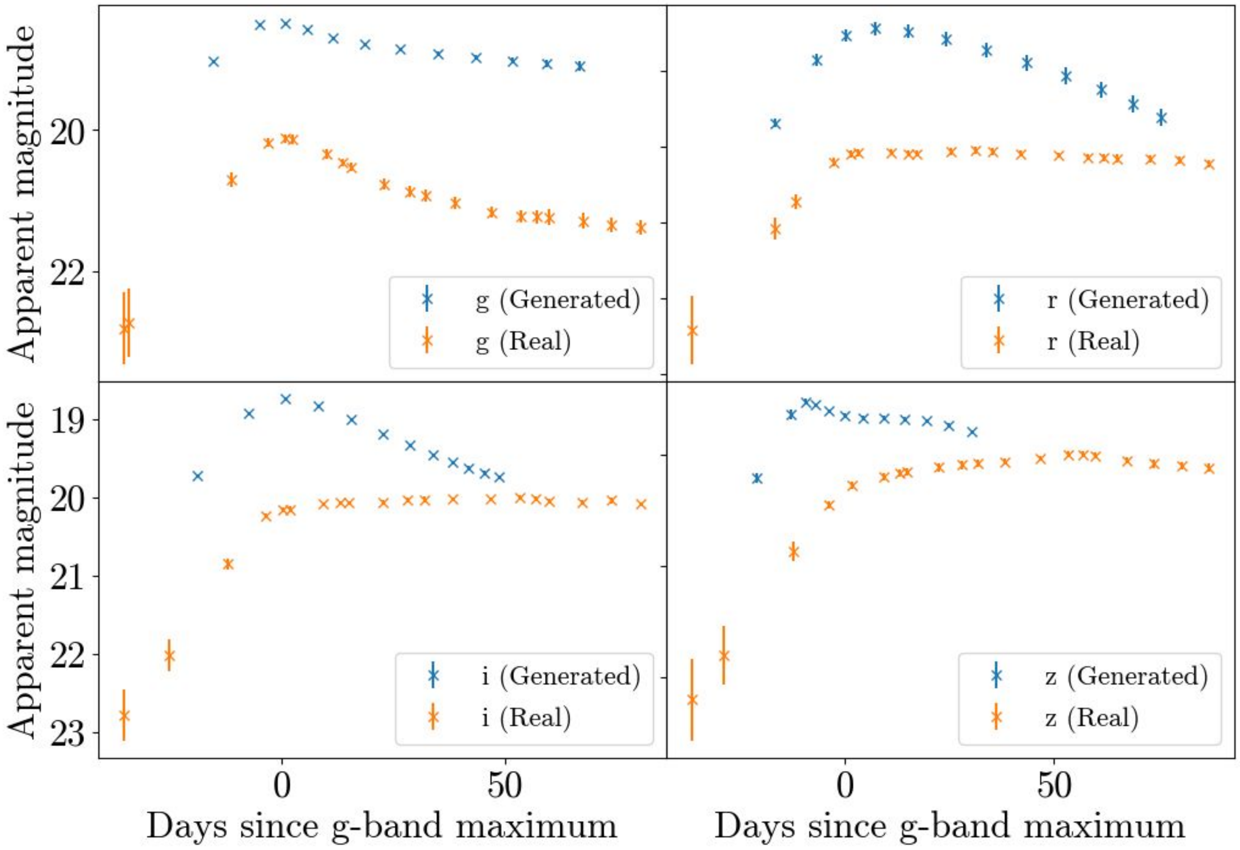
Training: Epoch 150



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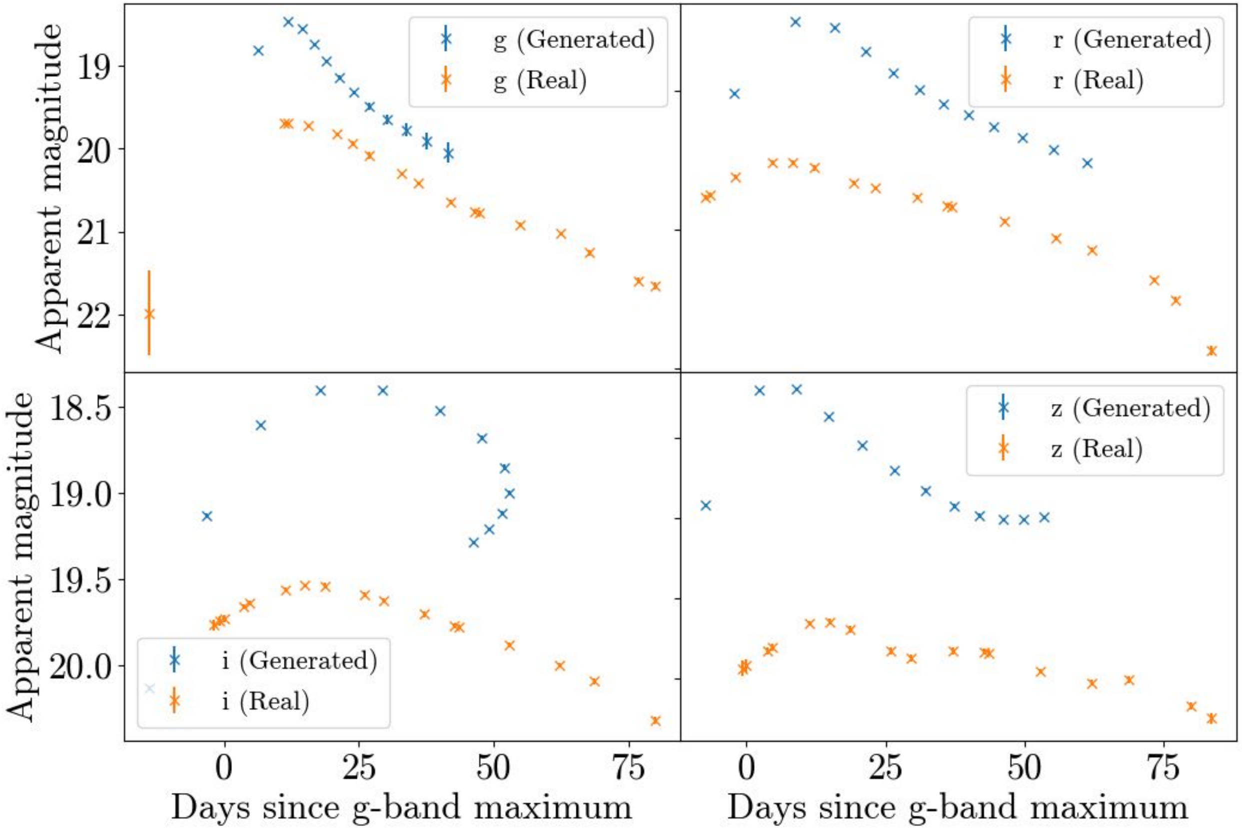
Training: Epoch 200



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Summary

- We present luminosity functions for core-collapse supernovae in DES, and compare supernova and host properties with samples from LOSS and ZTF
- Some evidence to suggest differences in the luminosity function between different surveys
    - Could indicate a redshift evolution in the luminosity function
    - Could also result from simpler effects such as differing levels of host extinction
- Unexplained difference in host U-R colour between DES and ZTF
    - Not redshift evolution, selection effects, systematic offset
- Ongoing work to generate synthetic supernovae to help improve photometric classification

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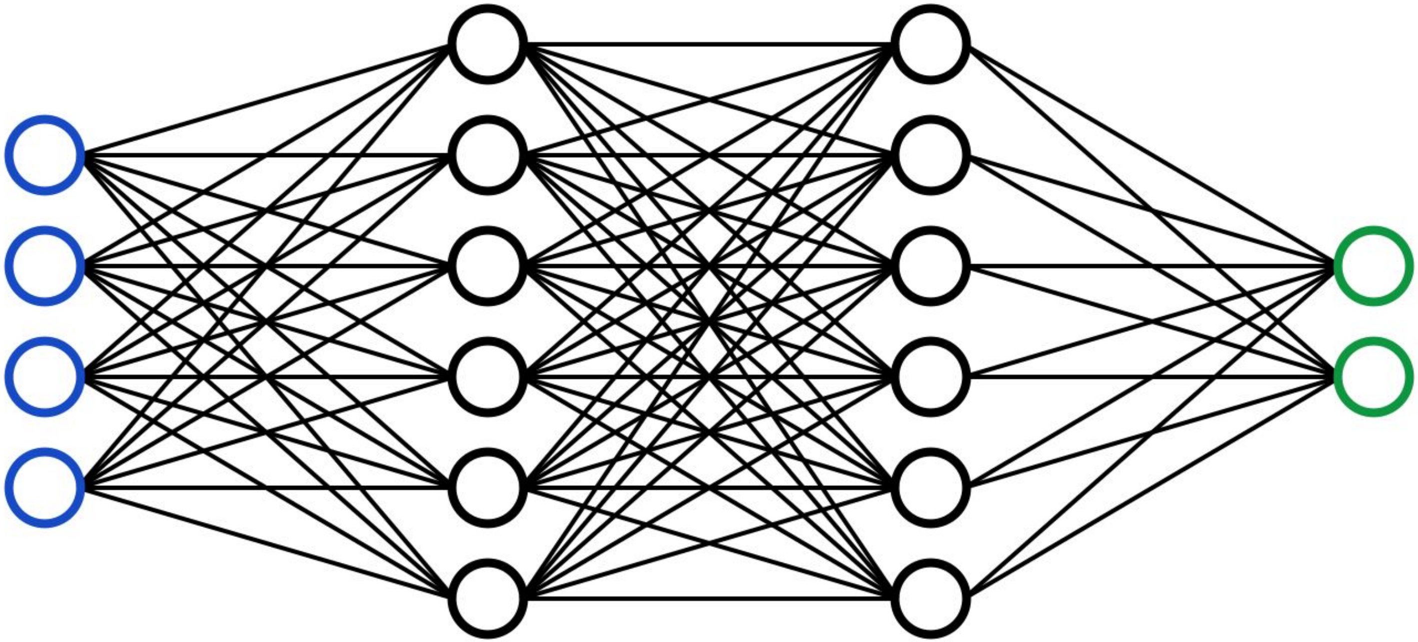
Neural Networks



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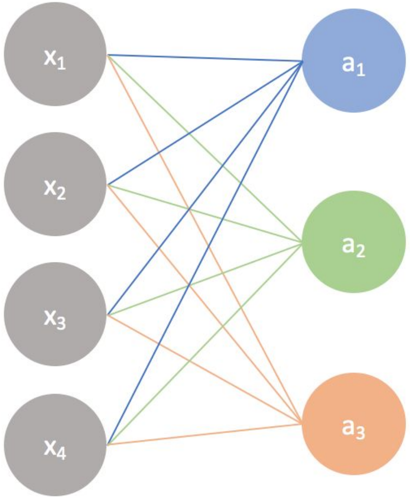
Neural Networks

Input layer Output layer



A simple neural network

$$\begin{bmatrix} w_1 & w_2 & w_3 & w_4 \\ w_1 & w_2 & w_3 & w_4 \\ w_1 & w_2 & w_3 & w_4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} b \\ b \\ b \end{bmatrix} = \begin{bmatrix} w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b \\ w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b \\ w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b \end{bmatrix} \xrightarrow[\textit{activation}]{} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

True probability distribution (one-hot)

$$H(p,q) = -\sum_{x \in \text{classes}} p(x) \log q(x)$$

Your model's predicted probability distribution

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Recurrent Neural Networks